MMM-A-182a
October 6, 1967
SUPERSEDING
Int Fed Spec MMM-A-00182(GSA-FSS)
November 29, 1962 and
MIL-C-12850(Cml C)
June 30, 1953
(See section 6)

FEDERAL SPECIFICATION

ADHESIVE, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1 SCOPE AND CLASSIFICATION

1 1 Scope. This specification covers a cold-bonding rubber adhesive for use with natural or synthetic rubber It is commercially known as rubber cement for cold-patching

1 2 Classification.

1.2.1 Types, The adhesive covered by this specification shall be of the following types (see 6 2):

Type I - Adhesive with a minimum of 6 percent rubber
Type II - Adhesive with 10 to 12 percent rubber

2. APPLICABLE DOCUMENTS

2 1 Specifications and standards The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

Federal Specifications

ZZ-I-550 - Inner Tube, Pneumatic Tire.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676 - Boxes, Setup
PPP-C-96 - Cans, Metal, 28 Gage and Lighter

Federal Standards

Fed Std No 123 - Marking for Domestic Shipment (Civilian Agencies)
Fed Test Method Std No 175 - Adhesives Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The index which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington, D C , 20402

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D C , Atlanta, Chicago, Kansas City, Mo , Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies

FSC 8040

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Military Specification

MIL-F-51109 - Faceblank, Protective Mask, C12R4
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

Federal Hazardous Substances Labeling Act·

Code of Federal Regulations, Title 21, Parographs 191.7 and 191.101.

American Society for Testing And Materials (ASTM) Standards·

D 86-62 - Distillation of Petroleum Products
D 297-61T - Chemical Analysis of Rubber Products.
D 412-64T - Tension Testing of Vulcanized Rubber.
D 573-53 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D 1266-64T - Sulfur in Petroleum Products and Liquified Petroleum (LP) Gases (Lamp Method)

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race St , Philadelphia, Pa., 19103,)

3 REQUIREMENTS

3.1 Materials

3 1 1 Rubber The rubber used in making the adhesive shall be first quality smoked sheet, pale crepe, or fine Para The rubber shall be milled or plasticized only enough to produce good brushing properties. Modifying resins may be added to the extent that the other requirements of this specification are met (see 4 2 1)

3 1 2 Solvent

3.1.2.1 Type I adhesive The solvent for type I adhesive shall be naphtha conforming to the distillation requirements in table I and shall be free of any sulfur or sulfur compound when tested as specified in 4 5.1 and 4 5 2.

3 1.2.2 Type II adhesive The solvent for type II adhesive shall be oil-free petroleum naphtha conforming to the distillation requirements in table I and shall be free of any sulfur or sulfur compound when tested as specified in 4 5 1 and 4 5.2

TABLE I. DISTILLATION REQUIREMENTS

	Initial boiling point (minimum)	End point (maximum)
*Naphtha	40°C. (104°F.)	143°C. (289°F.)

*The naphtha shall show a 98-percent recovery after having reached the end point

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3 2 Total nonvolatile content.

3.2 1 Type I adhesive The total nonvolatile content of type I adhesive shall be at least six-percent of the total weight of the adhesive when tested as specified in 4 5 3

3.2 2 Type II adhesive The total nonvolatile content of type II adhesive shall be no less than 12-percent of the total weight of the adhesive when tested as specified in 4 5 3.

3 2.3 Acetone extract of solid components The acetone extract of the solid components shall be not greater than 2.8 percent, by weight, of the adhesive when tested as specified in 4.5 4

3 3 Consistency. The adhesive shall spread easily with a spatula or brush and shall dry to a smooth uniform tacky film After evaporation of the solvent, the film produced shall lay where it is spread and not tend to draw itself together and, it shall be free of lumps, ridges, or streaks (see 4 5 5)

3.4 Adhesion

3 4 1 Type I adhesive. The adhesive shall be capable of producing sufficient adhesion over an area of 2 square inches so that no separation will occur with a load of 25 pounds when tested as specified in 4 5 6

3 4 2 Type II adhesive. The adhesive shall be capable of producing sufficient adhesion so that no separation will occur with a load of five pounds before and after aging (see 4.5.8), when tested as specified in 4.5 7

3.5 Workmanship. The finished adhesive shall be homogeneous, and free of lumps and foreign material.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the Inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to specified requirements

4 1.1 Objective evidence The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied

4 2 Lotting A lot shall consist of all the adhesive manufactured as one batch and submitted for delivery at one time

4.3 Sampling

4 3 1 For examination Sampling shall be conducted in accordance with MIL-STD-105.

4.4 Inspection

4 4 1 For examination. Sample containers of the adhesive, selected in accordance with 4 2 2.1, shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4 2 For tests The two pint samples, from each batch, shall both be tested as specified in 4 5 The acceptance number shall be zero

4 4 3 Classification of defects

(a) Adhesive, rubber

Categories	Defects
Critical	None defined
Major:	AQL 2 5 percent defective
101	Nonhomogeneous (3.5)
102	Lumps (3 5)
103	Foreign matter (3 5)

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(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1 5 percent defective
101	Quantity of fill incorrect
102	Shipping container not as specified
103	Quantity per shipping container not as specified
104	Closure of shipping container incorrect
105	Shipping container damaged
106	Leaking container
107	Marking of shipping container illegible, incorrect, incomplete

4 5 Tests Tests shall be performed as follows

4 5 1 Distillation Determine the distillation points of the solvents in accordance with ASTM D 86-62

4 5 2 Sulfur content Determine the sulfur content of the solvents in accordance with ASTM D 1266-64T

4 5 3 Total nonvolatile content Determine the nonvolatile content of the adhesive in accordance with Fed Test Msthod Std No 175

4 5.4 Acetone extract Evaporate the solvent from approximately 50 grams of adhesive, weighed to the nearest decigram, on a steam bath Use the residue for determining the acetone extract in accordance with ASTM D 297-61T. Calculate the percent acetone extract on the basis of the total adhesive by wlight.

4 5 5 Consistency Carefully examine the adhesive during the preparation of the test strips described in 4 5 6 for compliance with the consistency requirements in 3 3

4.5 6 Adhesion (type I adhesive) Preclean ten 1 by 6 inch strips of rubber conforming to ZZ-1-550 with fine sandpaper and then wipe them clean with a cloth moistened with petroleum ether Apply a coating of the adhesive cold to one-half of each strip with a spatula or brush and then allow to dry for 20 minutes. Prepare five specimens by placing the adhesive coated surfaces of five pair of strips together in such a way that the cemented area will be two square inches and press the joints firmly Allow the specimens to stand for 24 hours Clamp the unbonded ends of each specimen in the jaws of a testing machine specified in ASTM D 412-64T Operate the machine at a rate of 20 inches per minute Average the breaking force recorded on the 5 specimens.

4 5 7 Adhesion (type II adhesive) Prepare ten 1 by 6 by 0.1 inch strips from cured slabs of rubber conforming to MIL-F-51109 Buff one surface of each strip with No 80 emery cloth or a wire brush sufficiently to roughen the surface Wash the buffed surfaces with benzene and dry thoroughly. Apply one coat of adhesive to the buffed surface to within one inch of the end of each strip and dry for 15 minutes at 24° ± 3°C (75° ± 5°F). Apply a second coat of adhesive and repeat the drying operation. Assemble the strips in pairs with the coated surfaces in contact Place the assembled specimens on a smooth surface and pass a 2½ ± 1 ounce roller over each specimen five times using no pressure other than the weight of the roller Allow the bonded specimens to stand for 24 hours at 24° ± 3°C (75° ± 5°F.) before testing. Clamp the unbonded ends of each specimen in the jaws of a testing machine specified in ASTM D 412-64T Operate the machine at a rate of one inch per minute Average the breaking force recorded on the five specimens

4 5 8 Accelerated aging Prepare five specimens (see 4.5 7) and age them in an oven as specified in ASTM D 573-53 at 100° ± 2°C (212° ± 4°F) for 16 hours Remove the specimens from the oven, cool to room temperature, and test as specified in 4.5 7

5 PREPARATION FOR DELIVERY

5 1 Packaging The adhesove shall be packaged level A, B, or C, as specified by the contract or order (see 6 2)

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5 1 1 Level A The adhesive shall be packaged, as specified by the contract or order, in collapsible tubes of one ounce capacity, in cans of one pint or one allon capacity, or in five gallon capacity pails (see 6.2), in accordance with the applicable requirements of MIL-STD-290

5.1 1 1 Seals. Tubes, cans, and pails shall be furnished with inner seals which shall comply with and be applied in accordance with the requirements of MIL-STD-290

5 1 2 Level B, civil agencies The adhesive shall be packaged in accordance with table II.

5.1.3 Level C. Quantities of adhesive, as specified by the contract or order, shall be packaged so as to provide adequate protection against corrosion, deterieration, and damage from the supply source to the first receiving activity for immediate use The supplier may use his commercial practice, provided it fulfills the above requirements.

5.1.4 Intermediate packaging Packages of one ounce quantities of adhesive in collapsible tubes shall be intermediate packaged, level A or B, as specified by the contract or order (see 6.2)

5.1.4.1 Level A. One ounce collapsible tubes, packaged as specified in 5.1.1, shall be intermediate packaged in compliance with the requirements for level A packaging of tubes in MIL-STD-290.

5 1.4 2 Level B, civil agencies. The adhesive, packaged as specified in 5.1.2, shall be intermediate packaged in accordance with table II

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A, military. Adhesive, packaged as specified in 5.1.1 or intermediate packaged, as specified in 5.1 4.1, as applicable, shall be packed in accordance with the requirements for packing of compounds as specified in MIL-STD-290

TABLE II

PACKAGING AND PACKING--CONTAINERS--QUANTITIES

Packaging					Packing	
	Unit package		Intermediate package			
Size	Container	Quantity	Container	Quantity	Container	
1 oz	Tube, commercial type	12	PPP-B-566, PPP-B-665 or PPP-B-676 w/full height slotted interlocking partitions	4	PPP-B-636, class domestic, variety SW	
2 oz	Tube, commercial type	12	PPP-B-566, PPP-B-665 or PPP-B-676 w/full height slotted interlocking partitions	4	PPP-B-636, class domestic, variety SW	
8 oz	PPP-C-96, type V, class 2	12		6	PPP-B-636, class domestic, variety SW	
1 pt	PPP-C-96, type V, class 2			48	PPP-B-636, class domestic, variety SW	
1 gal	PPP-C-96, type V, class 2			4	PPP-B-636, class domestic, variety SW	

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5.2.2 Level B, civil agencies The adhesive shall be packed in accordance with table II

5 2 3 Level C The adhesive shall be packed to insure carrier acceptance and safe delivery to the first domestic destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportaion.

5 3 Marking

5 3 1 Civil agencies Unless otherwise specified in the contract or order, unit containers, intermediate packages, and shipping containers shall be marked in accordance with Fed Std. No 123 In addition, each unit container shall be marked, without using codes, to show the month and year of manufacture.

5 3 2 Military agencies In addition to any special marking required by the contract or order, unit containers, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 Each unit container shall be marked, without using codes, to show the month and year of manufacture

5.3 3 Precautionary marking Packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of the Federal Hazardous Substances Labeling Act and shall bear the following precautionary marking·

"POISON -- DANGER
Contains naphtha
Extremely flammable.
Vapor harmful.
Harmful if swallowed
Keep away from heat, sparks, and open flame.
Keep container closed when not in use.
Use with adequate ventilation
Avoid prolonged or repeated breathing of vapor
Avoid prolonged or repeated skin contact.
Keep out of reach of children.
First aid -- Call physician immediately
If swallowed -- Do not induce vomiting "

6 NOTES

6 1 Intended use. The type I adhesive covered by this specification is intended for use in cold patching of tire inner tubes and similar rubber articles. The type II adhesive covered by this specification is intended primarily for bonding components of natural rubber in the assembly of protective masks.

6 2 Ordering data Purchasers should select the preferred options permitted herein and include the following information in procurement documents·

(a) Title, number, and date of this specification.
(b) Type of adhesive required (see 1 2.1)
(c) Size of container required.
(d) Selection of applicable levels of packaging and packing required (see 5.1 and 5 2)
(e) Special marking if required (see 5 3.1 and 5 3.2)

6 3 Supersession data. Type I adhesive of this specification includes the requirements of type II, ZZ-C-191 Type II adhesive of this specification includes the requriements of MIL-C-12850

MILITARY INTEREST·

Custodian

Army - MU

Review activities·

Army - MR, MU, MD

User activity:

Army - SM

Preparing activity·

Army - MU (EA)

CIVIL AGENCY INTEREST·

Interested activity

VA-DMS

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